

Timeline of Technology: Final Technical Paper

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ABSTRACT

Timeline of Technology is the mature version of the project. This 3D narrative-driven exploration game integrates environmental storytelling, artifact-based interaction, and system-driven progression into a cohesive player experience. Expanding from its early prototype, the final version introduces a complete gameplay loop centered on exploration, artifact discovery, and multi-stage spacecraft repair within a museum of human technological history.

Set in a post-collapse Earth, players navigate curated environments representing key eras of technological advancement while uncovering the consequences of human innovation. Core systems, including the A.R.C.H.I.V.E wrist device, the evolving AI companion ECHO, and modular repair mechanics, work together to merge narrative delivery with gameplay progression. Each artifact serves both as an interactive mechanic and a storytelling medium, allowing players to reconstruct humanity's past while gradually restoring their means of escape.

The final implementation demonstrates a robust integration of gameplay systems, including environmental puzzles, artifact-based interactions, and responsive UI frameworks. The project highlights how interactive systems can convey complex themes such as technological dependency, and environmental collapse through a cohesive and interactive player experience.

1. INTRODUCTION

Timeline of Technology is a 3D exploration-based interactive experience in which players assume the role of a crash-landed traveller navigating the ruins of Earth, primarily within the *Museum of Human*

Achievement. The player's objective is to recover technological artifacts from different ears and use them to repair a damaged spacecraft, gradually restoring both functionality and narrative context.

In its final form, the project evolves beyond a proof-of-concept into a cohesive and immersive experience that tightly integrates gameplay mechanics with storytelling. The core gameplay loop consists of three interconnected systems:

1. **Exploration:** Player traverses a curated environment representing industrial, computational, and AI-driven ears, each designed with distinct visual language and spatial storytelling.
2. **Collection:** Interactive artifacts are discovered, scanned, and contextualized through the A.R.C.H.I.V.E device, revealing fragments of historical simulations and clues.
3. **Repair:** Collected artifacts are used within ship-based repair systems, triggering interactive minigames that restore ship components and unlock new stages and areas of gameplay.

A key design principle of the project is player-driven discovery. Rather than relying on explicit instruction, the game uses environment cues, artifact placement, and dialogue system feedback to guide player understanding of progression. This approach allows narrative and gameplay to emerge simultaneously, encouraging players to interpret the world through interaction.

By combining narrative progression with mechanical systems, *Timeline of Technology* positions itself within contemporary exploration-based game design, emphasizing agency, immersion, and the meaningful relationship between technology and human history.

2. RELATED WORKS

2.1 The Matrix: Interpreting Hidden Layers of Technology

The Matrix (Wachowski and Wachowski, 1999) inspired the project's interest in environments that contain information beneath the surface. The film uses symbolic visuals and coded interfaces to show how reality can be revealed through interaction with technology. The prototype reflects this idea by allowing players to scan and inspect objects in order to understand how they fit into the world. While the film focuses on philosophical themes, the prototype uses similar ideas in a more grounded and practical way.

2.2 Outer Wilds: Curiosity-Driven Discovery

Outer Wilds (Mobius Digital, 2019) is a strong example of how players can learn by observing and experimenting in the environment. The game encourages investigation through scattered clues and ancient technology. The prototype draws from this sense of curiosity by letting players inspect artifacts and uncover small pieces of information. While Outer Wilds presents a vast interconnected mystery, the prototype is currently focused on the basic systems that will support this style of discovery later.

2.3 Terminator: AI, Risk and Consequences

The Terminator series (Cameron, 1984) contributed a conceptual influence through its portrayal of artificial intelligence as a potential danger. The films explore how autonomous machines and AI could evolve beyond human control, shaping the environment in unforeseen and threatening ways. This influence informs the prototype's approach to technology by highlighting the stakes of interacting with intelligent systems. Players are encouraged to think critically about the risks and consequences inherent in advanced technology, even within a controlled exploratory setting.

2.4 Star Wars: Technology as Part of the World's History

The Star Wars universe (Lucas, 1977) presents technology as an integral part of its world, giving machines personality, history, and cultural significance. Devices and vehicles are not just functional tools—they carry meaning and reflect the societies that created them. This approach informs the prototype's design of artifacts, encouraging each object to feel intentional and narratively resonant. By embedding history and context into technological objects, the prototype supports discovery and learning, allowing players to infer aspects of the world's past and culture through interaction.

2.5 Subnautica: Environmental Discovery and Player Agency

Subnautica (Unknown Worlds Entertainment, 2018) illustrates how an immersive environment can encourage exploration and problem-solving through observation and interaction. Players uncover the world's story primarily through its ecosystem and technology. The prototype applies this approach by emphasizing curiosity and gradual discovery, allowing objects and artifacts to reveal information through inspection.

2.6 Combined Perspective

Across these works, a shared idea emerges: technology can carry history, emotion, atmosphere, and story. The Matrix reveals hidden systems. Star Wars presents machines as cultural artifacts. Outer Wilds treats devices as clues. Terminator underscores the risks and consequences of advanced AI. Subnautica

highlights learning through interaction. The prototype follows this same philosophy. Even in its early, interaction-focused stage, its systems are designed so that artifacts guide the player's understanding of the environment. They also establish the foundation for deeper narrative elements in the future.

3. DESCRIPTION OF FINAL PROJECT DEVELOPMENT PROCESS

3.1 Final Project Overview

The final version of *Timeline of Technology* represents a transition from isolated system development to a fully integrated interactive experience. The gameplay loop has been refined to ensure consistency across systems, allowing players to progress naturally through exploration, collection, and repair.

Players begin in a damaged spacecraft and are encouraged to explore the surrounding museum environment. Artifacts are distributed strategically to guide player movement and encourage interaction. Each artifact contributes to gameplay progression by enabling repair tasks.

3.2 Visual Documentation of Final Project

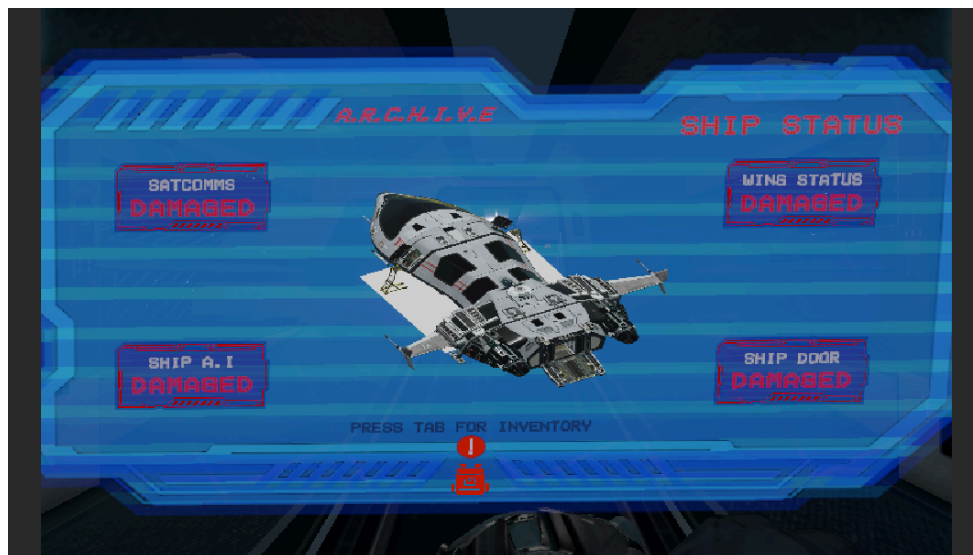


Fig. 1. An image of the ship's status in the Archive UI, showing all the damages that need to be fixed.

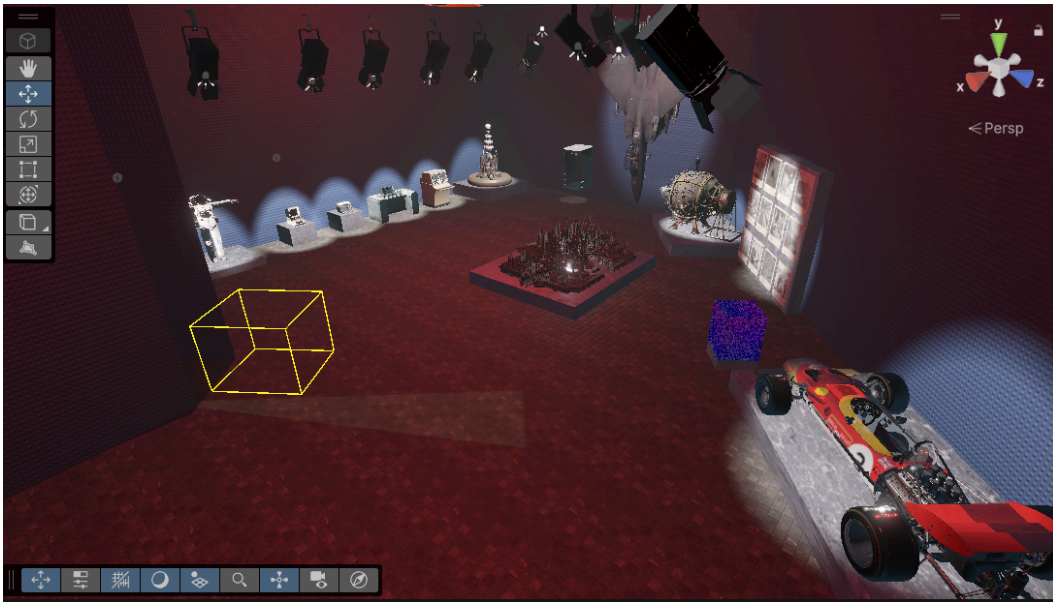


Fig. 2. An image showing the interior of room 2, where players must try and retrieve a microchip inside the blue shield (on the right).

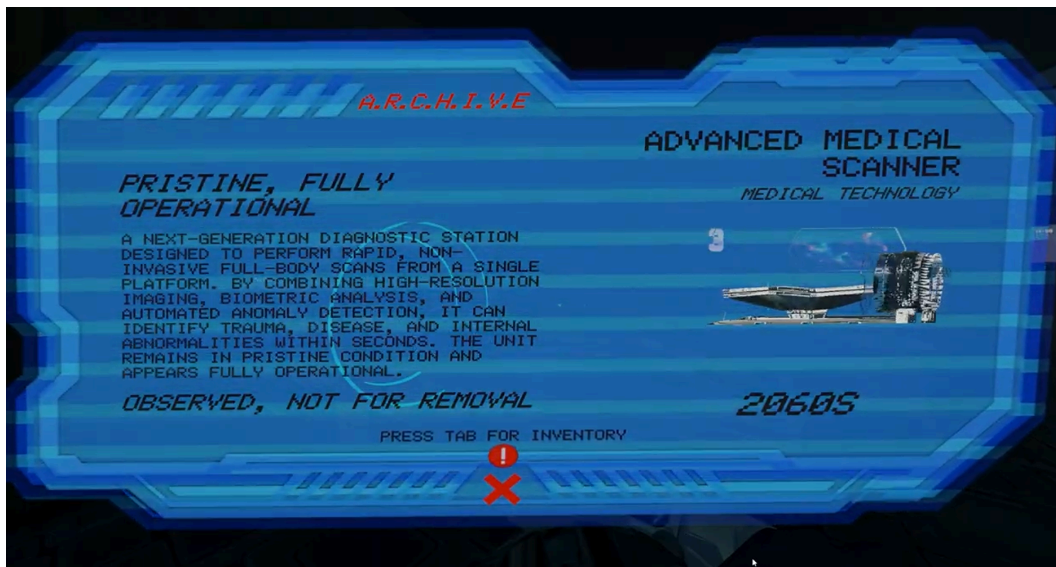


Fig. 3. An image showing a viewable artifact, with its condition, name, description and when it was made

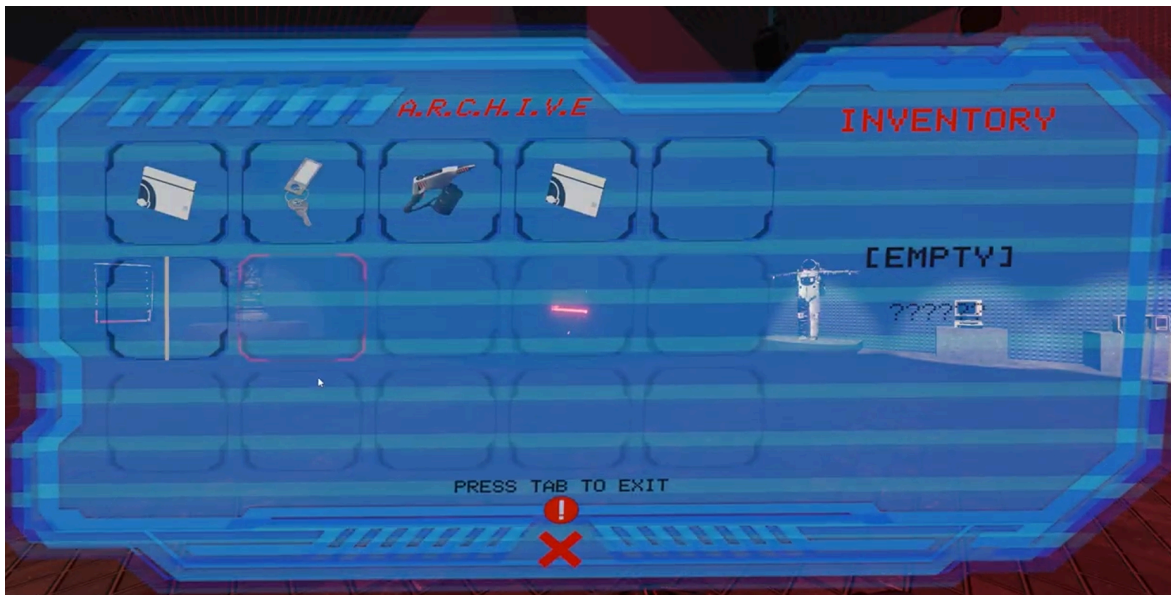


Fig. 4. An image showing the inventory, where there are slots for each of the items the player picks up. Each slot can be clicked on to reveal more information.



Fig. 5. An image showing the laser that the player is reflecting using advanced reflective material in order to access a computer chip in the second room needed for repairs (see Fig.1)



Fig 6. An image showing the player using the welding tool, where they must attach a metal sheet that they cut out to the ship. The tool malfunctions at random intervals, making the player pay attention to when that might happen and choose when to weld and when not to weld.

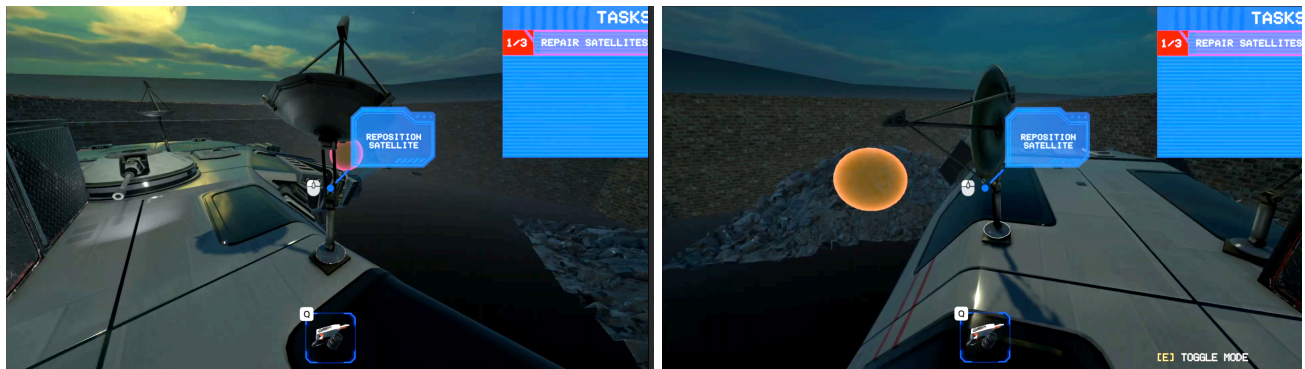


Fig 7. An image showing the satellite reposition game, where the player has to control the rotation and swivel the different joints to align its antenna into the orangish-red sphere

3.3 Design and Development Process

Development followed an iterative process, beginning with the implementation of individual systems and progressing toward full integration. Early stages focused on core mechanics such as object interaction and inventory management. These systems were refined through testing to improve

responsiveness and usability.

As development progressed, integration challenges emerged. These included input conflicts, inconsistent UI behaviour, and system communication issues. These problems were addressed through code restructuring and the adoption of a modular system design.

Team collaboration required coordination across multiple disciplines, including programming, art, and design. Clear division of responsibilities and iterative testing were essential to maintaining project stability.

There were also implementation complications that arose as well. The majority was caused by merging conflicts, as environmental changes could be very hard to integrate seamlessly. We were able to fix this by having more frequent commits on GitHub to avoid big complications. Implementation of animation also caused some difficulties, as certain parts would not react properly to others. So it was decided that the use of Blender would be better, especially as we had to animate and render clips for our project for the beginning and ending sequences

3.3.1 *Third-Party Assets.* The project incorporates third-party resources to support development efficiency:

- Unity Asset Store and Sketchfab models for environmental assets
- Audio libraries for interaction feedback
- UI frameworks for inventory systems

These assets were modified and integrated to align with the project's visual and functional goals.

3.4 Interaction Scenarios

3.4.1 *Exploration Scenarios.* The player navigates the museum environment using spatial cues such as lighting, layout, and object placement to guide exploration. Environmental storytelling elements, including scattered journals and contextual details, provide narrative insight and subtly direct player attention toward key areas.

3.4.2 *Artifact Interaction Scenarios.* When approaching an interactable object, a highlight effect

signals availability. Upon interaction, there are 2 different ways they can be used. The first is that the player triggers a pickup sequence with visual and audio feedback, and the object is added to the inventory. Then, A.R.C.H.I.V.E. UI gives them information on what it's for, when it was made and what it is. They are known as interactables. The next type of artifact interaction objects are called viewables and are meant to be for narrative purposes. A pop-up of the item's description will show up on the player's screen, where they can read key information they may need to help them progress throughout the story.

3.4.3 *Inventory and Journal Scenarios.* The inventory system allows players to review collected artifacts and track progression through a structured UI. Integrated journals and logs provide additional context, offering narrative details and gameplay hints that support player understanding without interrupting flow.

3.4.4 *Dialogue and Guidance Scenarios.* Dialogue is triggered contextually through interactions or progression events. It provides subtle guidance and reinforces objectives without being overly directive, complementing environmental cues and UI feedback to support player navigation and decision-making.

3.4.5 *Repair Scenarios.* Players return to the spacecraft to use collected artifacts in repair stations, initiating minigames such as gear alignment, welding, or laser-based reflection and redirection. These interactions require spatial reasoning, precision, or continuous input. Successful completion updates the ship's status and reinforces progression within the gameplay loop. Unsuccessful attempts trigger helpful cues to correct the player on how to finish the repair sequence

3.5 Interaction Diagrams

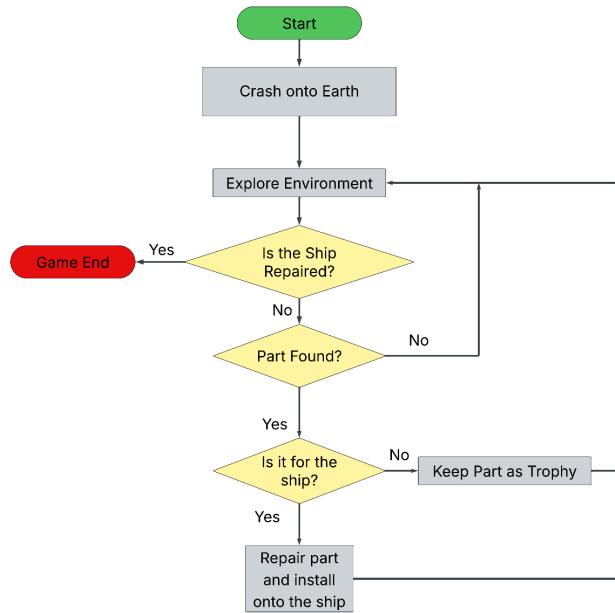


Fig. 3. Player interaction flows from exploration to repair progression.

3.6 System Diagrams

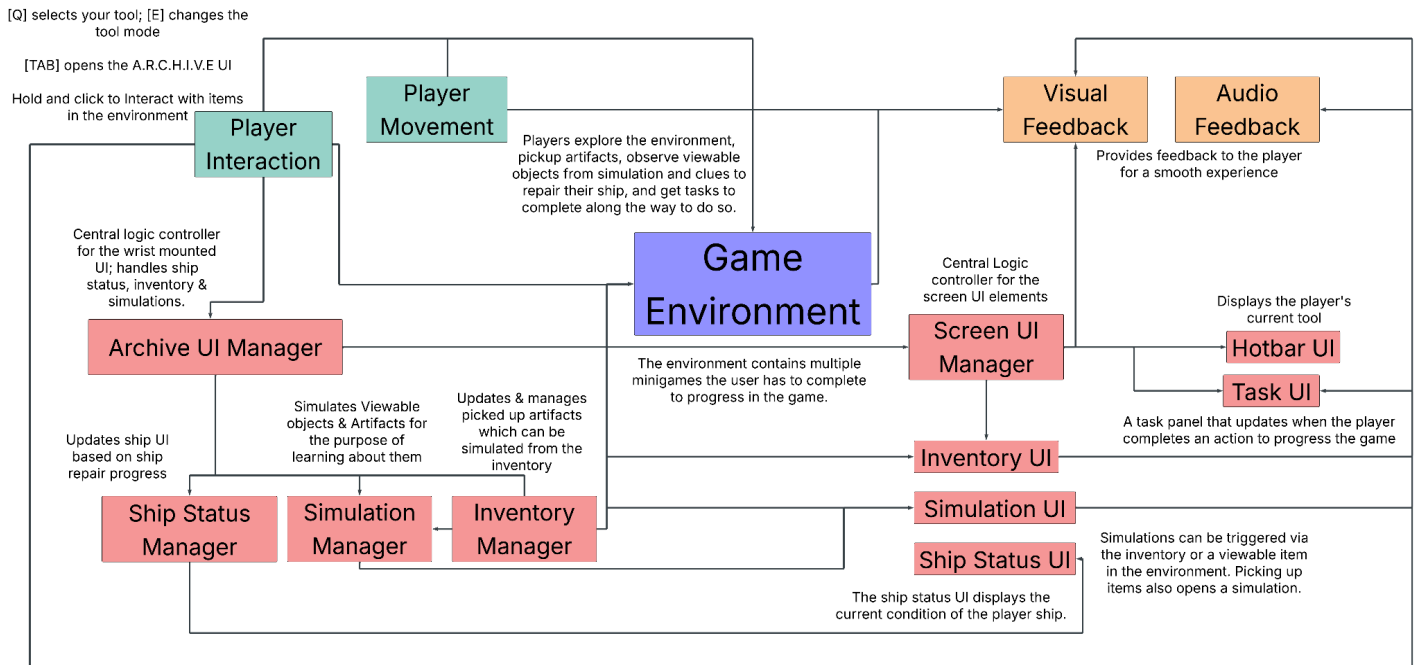


Fig. 4. System architecture showing relationships between gameplay systems.

4. CONCLUSION

The final version of our project demonstrates the successful transformation of an interaction-focused prototype into a cohesive and functional interactive experience. Through iterative development and system integration, the project evolved from isolated technical experiments into a unified gameplay system that combines exploration, artifact collection, and repair-based mechanics. The implementation of multiple interconnected systems, including object interaction, inventory management, UI feedback, and multi-stage repair minigames, reflects a comprehensive approach to both technical design and player experience.

A key contribution of the project lies in its emphasis on player-driven discovery. Rather than relying on explicit instruction, the system leverages environmental design, visual feedback, and interaction cues to guide player behaviour. This approach aligns with contemporary design practices in exploration-based games, where player agency and interpretation play a central role in engagement. The inclusion of varied repair minigames further enhances this experience by introducing mechanical diversity, ensuring that core gameplay remains engaging while reinforcing the thematic concept of learning about different forms of technology.

From a technical perspective, the project highlights the importance of modular system design and iterative refinement. Challenges related to input handling, UI clarity, and system integration required restructuring and optimization of the codebase. Addressing these issues not only improved system stability but also contributed to a more consistent and responsive user experience. The development process demonstrated the value of collaboration, clear task allocation, and continuous testing in managing a complex interactive project.

Despite achieving its primary objectives, the project also presents opportunities for future expansion. Potential improvements include the introduction of more complex narrative elements, additional environments, and expanded interaction systems that further deepen player engagement. Enhancing the visual fidelity of assets, refining animation systems, and integrating more advanced progression mechanics could also contribute to a more immersive experience.

In conclusion, *Timeline of Technology* establishes a strong foundation for continued development by

successfully integrating core gameplay systems within a coherent design framework. The project not only demonstrates technical feasibility but also reflects an understanding of interactive design principles, particularly in relation to player engagement, system feedback, and environmental storytelling. As such, it serves as both a completed project and a scalable platform for future iteration and expansion.

5. REFERENCES

- [1] J. Cameron. 1984. The Terminator. Orion Pictures.
- [2] G. Lucas. 1977. Star Wars: Episode IV A New Hope. Lucasfilm.
- [3] Mobius Digital. 2019. Outer Wilds. Annapurna Interactive.
- [4] Unknown Worlds Entertainment. 2018. Subnautica. Unknown Worlds Entertainment.
- [5] L. Wachowski and L. Wachowski. 1999. The Matrix. Warner Bros.

6. APPENDIX

6.1 Task Allocation and Team Contributions

This project was developed collaboratively, with each team member contributing to specific areas of design, development, and implementation. The following outlines the detailed responsibilities of each member throughout the final production of *Timeline of Technology*.

6.1.1 Max (Maksim Pralat). Max served as the project lead, overseeing production and coordinating the overall development process. He managed timelines, organized team workflows, and ensured that milestones were met. In addition, he contributed to programming refinements and supported the integration of core gameplay systems.

6.1.2 Zed (Chizimuzo Ikejiani). Zed was the programming lead, responsible for system architecture and overall code structure. He ensured that different systems were properly integrated and maintained consistency across the codebase. He also managed version control workflows and coordinated the integration of team contributions.

6.1.3 Santiago Bucio-Cano. Santiago worked as the visual developer, focusing on environment design and overall visual presentation. He was responsible for constructing and refining the game environment, as well as creating animations that enhanced immersion and supported the player experience.

6.1.4 Phoebe (Phuong Nguyen). Phoebe was responsible for 3D modeling and support contributed to

the visual identity of the project. She refined key assets used throughout the environment, ensuring consistency in style and alignment with the game's theme.

6.1.5 Ehsan Karimzada. Ehsan contributed to programming, with a focus on implementing dialogue systems and sound effects. His work supported both gameplay functionality and the overall atmosphere of the game, enhancing player immersion. He also supported debugging and optimization to ensure system stability.

6.2 Team Collaboration

All members contributed collectively to game design decisions, narrative development, and playtesting. The team worked collaboratively to refine gameplay flow, ensure consistency across systems, and maintain a cohesive player experience throughout development.

Timeline of Technology: Asset Reference & Attribution

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SANTIAGO BUCIO-CANO, 219029677

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Font

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- Typodermic Fonts. (2025). *Ethnocentric* [Font]. DaFont.
- License: Free for personal use only (Typodermic Fonts license).

UI Elements

Asset: Free Outline Shader

- Source: Unity Asset Store
<https://assetstore.unity.com/packages/vfx/shaders/free-outline-326925>
- Ameye, A. (2026). *Free Outline* [Unity shader asset]. Unity Asset Store.
- License: Standard Unity Asset Store License.
<https://unity.com/legal/as-terms>

Asset: Sci-Fi UI Cells Pack

- Source: [itch.io](https://aniversal.itch.io/sci-fi-ui-cells-pack)
<https://aniversal.itch.io/sci-fi-ui-cells-pack>
- aniversal. (2025). *Sci-fi UI cells pack* [UI asset]. [itch.io](https://aniversal.itch.io).
- License: Free for commercial and personal use (custom itch.io license by creator).

Asset: Sci-Fi UI Border (Blue Interface)

- Source: PNGTree
https://pngtree.com/freepng/technology-border-ui-user-interface-blue-sci-fi-screen-index-bar_6594009.html
- PNGTree. (n.d.). *Technology border UI user interface blue sci-fi screen index bar* [PNG image]. PNGTree.
- License: PNGTree Free License.

Asset: Sci-Fi UI Border (Game Interface)

- Source: PNGTree
https://pngtree.com/freepng/video-game-technology-border-ui-user-interface-sci-fi-screen-index-bar_6594000.html
- PNGTree. (n.d.). *Video game technology border UI user interface sci-fi screen index bar* [PNG image]. PNGTree.
- License: PNGTree Free License.

Asset: X Mark Icon

- Source: IconsDB
<https://www.iconsdb.com/red-icons/x-mark-icon.html>
- IconsDB. (n.d.). *X mark icon* [Icon]. IconsDB.
- License: Free for personal and commercial use.

Asset: UI Keybind Icons Pack

- Source: Fab
<https://www.fab.com/listings/cfaab761-d03b-4207-8a07-a5dcc23cc776>
- MengGe_KKD. (2026). *UI Keybind Icons Pack* [UI asset]. Fab.
- License: Standard License (Fab).
<https://www.fab.com/eula>

Sound Effects

Assets: Various Sound Effects and Music

- Source: Pixabay
- Links:
<https://pixabay.com/sound-effects/3-up-1-89190/>
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- Pixabay. (n.d.). *Various sound effects and music* [Audio]. Pixabay. <https://pixabay.com/>
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Asset: Sound Effect

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<https://freesound.org/people/RescopicSound/sounds/751756/>
- RescopicSound. (n.d.). *Sound effect* [Audio]. Freesound.
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Assets: Keyboard, Mouse, Switch Sound Effects

- Source: Mixkit
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<https://mixkit.co/free-sound-effects/keyboard/>
<https://mixkit.co/free-sound-effects/mouse/>
<https://mixkit.co/free-sound-effects/discover/switch/>
- Mixkit. (n.d.). *Sound effects* [Audio]. Mixkit. <https://mixkit.co/>
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<https://mixkit.co/license/>

Character

Asset: Cyberpunk Bounty Hunter Character

- Source: CGTrader
<https://www.cgtrader.com/free-3d-models/character/sci-fi-character/cyberpunk-bounty-hunter-character-low-and-high-poly-both>

- artistofthefmonth. (2026). *Cyberpunk Bounty Hunter character LOW AND HIGH POLY both free low-poly 3D model* [3D model]. CGTrader.
- License: Royalty-Free License (CGTrader)

Asset: Cyber Meta-Human Head (#PortraitFromTheFuture)

- Source: Sketchfab
<https://sketchfab.com/3d-models/cyber-meta-human-head-portraitfromthefuture-683e2fd4850d49268c5363ca0b1e89ad>
- Léonard Doye (Leoskatemanpro). (2021). *Cyber Meta-Human Head #PortraitFromTheFuture* [3D model]. Sketchfab.
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Asset: The Division Inspired Agent Watch

- Source: Sketchfab
<https://sketchfab.com/3d-models/the-division-inspired-agent-watch-2607bd6a758347e3b6f689fffee93b23>
- Rens_Hoedeman. (2021). *The Division Inspired Agent Watch* [3D Model]. Sketchfab
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Interactable Artifacts

Asset: Nineteen Eighty-Four (1984) by George Orwell

- Source: Sketchfab
<https://sketchfab.com/3d-models/nineteen-eighty-four-1984-by-george-orwell-5ee608c32a8c4e5aa4034405c7e63a85>
- byamazaky. (2022). *Nineteen Eighty-Four (1984) by George Orwell* [3D model]. Sketchfab.
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Asset: Futuristic Welding Tool

- Source: Sketchfab
<https://sketchfab.com/3d-models/futuristic-welding-tool-133d999bafc24bfd8d46329eaf7f3d08>
- Yorus01. (2023). *Futuristic Welding Tool* [3D model]. Sketchfab.
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Asset: Chip Module

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<https://sketchfab.com/3d-models/lcd-16x2-067c99a33b274263b611bdf06f1b3c8a>
- Zeyad Hamed. (2024). *LCD 16x2* [3D model]. Sketchfab.
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<https://creativecommons.org/licenses/by/4.0/>

Asset: Circuit Board

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- Philip Osborne. (2018). *Circuit Board* [3D model]. Sketchfab.
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Asset: Satellite Modul

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<https://sketchfab.com/3d-models/satellite-modul-bae8489c6f2a4a9cb252cd6082ef9f81>
- Multipainkiller Studio. (2018). *Satellite Modul* [3D model]. Sketchfab.
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Asset: Keycard + Scanner

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- Yadira. (2025). *Keycard + Scanner* [3D model]. Sketchfab.
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Asset: Electric Door Key - 5MB

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- Mehdi Shahsavan. (2024). *Electric Door Key - 5MB* [3D model]. Sketchfab.
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Asset: Industrial Electrical Box

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<https://sketchfab.com/3d-models/industrial-electrical-box-de13c96dad014bf098df03030f314fee>
- Krystofeles. (2025). *Industrial Electrical Box* [3D model]. Sketchfab.

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Spaceship

Asset: Futuristic Spaceship

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<https://sketchfab.com/3d-models/futuristic-spaceship-89b1b3f200c84d5a990ae95c0e302848>
- dannzjs. (2024). *Futuristic Spaceship* [3D model]. Sketchfab.
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<https://www.fab.com/listings/164970c5-f9aa-4cbd-a18b-3ef9fab2edf0>
- Bruno Leal 3D. (2022). *Sci-Fi Floor Panel* [3D asset]. Fab.
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Asset: Satellite dishes

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<https://sketchfab.com/3d-models/satellite-dishs-b6885ee2eccc43929785dd6966e036a5>
- 3DHaupt. (2013). *Satellite dishs* [3D model]. Sketchfab.
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<https://sketchfab.com/3d-models/exit-button-switch-b97fb8b1b8cc4b008877a0161d9ee26c>
- YouniqueIdeaStudio. (2023). *Exit Button Switch* [3D model]. Sketchfab.
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- Shutterstock. (2021). *SYSTEM FAILURE error message flashing on screen animation* [Stock video; ID: 1072709171]. Shutterstock.

- License: Licensed via Shutterstock (standard stock license).

Asset: Warning Sign Alert Red

- Source: YouTube
<https://www.youtube.com/watch?v=GOjBV0DtQuY>
- MSH STUDIOS. (2023). *WARNING SIGN ALERT RED - VIDEO, GRAPHIC EFFECTS #3 #free #nocopyright* [Video]. YouTube.

Rooms' Props

Asset: Double Pipe Straight

- Source: Sketchfab
<https://sketchfab.com/3d-models/double-pipe-straight-0278e756bf824b60b66265be791e461a>
- TUGBOAT GAMES. (2021). *Double Pipe Straight* [3D model]. Sketchfab.
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Asset: Old Industrial Pipe Pack (PBR)

- Source: Sketchfab
<https://sketchfab.com/3d-models/old-industrial-pipe-pack-pbr-a10a6011334e4d11a7023d67705ac1d4>
- SLASH / RENDAR. (2020). *Old Industrial Pipe Pack (PBR)* [3D model]. Sketchfab.
- License: Creative Commons Attribution (CC BY 4.0).
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Asset: Industrial Wall Light

- Source: Sketchfab
<https://sketchfab.com/3d-models/industrial-wall-light-509092cf10b742e3a8d026c929470289>
- Haytham918. (2023). *Industrial Wall Light* [3D model]. Sketchfab.
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Asset: 1930s Heavy Industrial Wall Lamps

- Source: Sketchfab
<https://sketchfab.com/3d-models/1930s-heavy-industrial-wall-lamps-b63c31b9c6a04284b665af8a0a93d456>
- Mad_Lobster_Workshop. (2023). *1930s Heavy Industrial Wall Lamps* [3D model]. Sketchfab.

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Asset: Vintage Industrial Control Panels

- Source: Sketchfab
<https://sketchfab.com/3d-models/vintage-industrial-control-panels-fa873b30cd174dbb9f77ada367ae1780>
- Mad_Lobster_Workshop. (2023). *Vintage Industrial Control Panels* [3D model]. Sketchfab.
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Asset: Electrical Panel

- Source: Sketchfab
<https://sketchfab.com/3d-models/electrical-panel-75367f3627374636ad1d9ad858a66c6b>
- Prstk. (2024). *Electrical Panel* [3D model]. Sketchfab.
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Asset: Stage Lights Barn Doors

- Source: Sketchfab
<https://sketchfab.com/3d-models/stage-lights-barn-doors-10d9247f691d47158751b8c14971bfcc>
- Rowley, M. (2016). *Stage Lights Barn Doors* [3D model]. Sketchfab.
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Asset: Low poly Satellite

- Source: Sketchfab
<https://sketchfab.com/3d-models/low-poly-satellite-2ae74b35f4b7448084ff7572c756e922>
- Green fish. (2022). *Low poly Satellite* [3D model]. Sketchfab.
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Asset: Industrial Toolbox

- Source: Sketchfab
<https://sketchfab.com/3d-models/industrial-toolbox-5037b5bea4bf4bd7bf53d581d478bd90>
- LuddePudde. (2019). *Industrial Toolbox* [3D model]. Sketchfab.

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Asset: Key with Tag

- Source: Sketchfab
<https://sketchfab.com/3d-models/key-with-tag-16ea3fbecc6346df9859f0e18406951b->
- AxonDesigns.(2022). *Key with Tag* [3D model]. Sketchfab.
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Asset: Industrial Electricity Lever

- Source: Sketchfab
<https://sketchfab.com/3d-models/industrial-electricitys-lever-66915915e3234d319415d6bfb4e3a59e>
- Amine.Elouneq. (2021). *Industrial Electricity's Lever* [3D model]. Sketchfab.
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Asset: Game Locker

- Source: Sketchfab
<https://sketchfab.com/3d-models/game-locker-3c2bea736afa481aa2e74a568655b831>
- amanpatel9725. (2025). *Game Locker* [3D model]. Sketchfab.
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Asset: Industrial Air Conditioner

- Source: Sketchfab
<https://sketchfab.com/3d-models/industrial-air-conditioner-82797367c2b94fb1981e556577a3c898>
- HusarL. (2022). *Industrial Air Conditioner* [3D model]. Sketchfab.
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Asset: Desk Lathe

- Source: Sketchfab
<https://sketchfab.com/3d-models/desk-lathe-7b7ef0cdefdf4a0d9e483a026af98fd2>
- Shavin, S. (2020). *Desk Lathe* [3D model]
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Asset: Lathe Machine

- Source: Sketchfab
<https://sketchfab.com/3d-models/lathe-machine-ff2240fac41d42fc84eef90a0e241825>
- ridhobahri. (2024). *Lathe Machine* [3D model]. Sketchfab.
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Asset: Rusted Industrial Water Tank

- Source: Sketchfab
<https://sketchfab.com/3d-models/rusted-industrial-water-tank-cfc37d0aff454f29a27c8ab3325f445b>
- Arghanion's PuzzleBox. (2019). *Rusted Industrial Water Tank* [3D model]. Sketchfab.
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Asset: Tanker Box Metallic Industrial

- Source: Sketchfab
<https://sketchfab.com/3d-models/tanker-box-metallic-industrial-6fef574d084c46f5a95cea106d1bb75b>
- Mehdi Shahsavan. (2024). *Tanker Box Metallic Industrial* [3D model]. Sketchfab.
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Asset: Laundry Machine

- Source: Sketchfab
<https://sketchfab.com/3d-models/hl2-inspired-laundry-machine-1-11b2bf7ec7824a32b5fe25e2c7eefd42>
- Kidney_02. (2021). *HL2 Inspired Laundry Machine* [3D model]. Sketchfab.
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Asset: Auto Turret Machine Gun

- Source: Sketchfab
<https://sketchfab.com/3d-models/auto-turret-machine-gun-1f186169b31d44c7bdf874f97508b340>
- NghiaNguyeen. (2022). *Auto Turret Machine Gun* [3D model]. Sketchfab.
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Asset: Egg

- Source: Sketchfab
<https://sketchfab.com/3d-models/an-eggweekly-challenge-18371155b1074ecea96ead2d269aaae0>
- Zeyad Hamed. (2022). *Egg (Weekly Challenge)* [3D model]. Sketchfab.
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Asset: Low Poly Sci-Fi Containers Pack

- Source: Sketchfab
<https://sketchfab.com/3d-models/lowpoly-scifi-containers-pack-b885c4163d6f458ba484f55149fdf8a3>
- IrisProcess. (2023). *Low Poly Sci-Fi Containers Pack* [3D model]. Sketchfab.
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Asset: Sci-Fi Windows

- Source: Sketchfab
<https://sketchfab.com/3d-models/scifi-windows-0ebb0cc000c74324a0a0f622fcb3bc39>
- Sousinho. (2023). *Sci-Fi Windows* [3D model]. Sketchfab.
- License: Free Standard License (Sketchfab).
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Asset: Sci-Fi Drone

- Source: Sketchfab
<https://sketchfab.com/3d-models/scifi-drone-e5efecf43ca74d2898a9053bd51f7f76>
- Badge. (2024). *Sci-Fi Drone* [3D model]. Sketchfab.
- License: Free Standard License (Sketchfab).
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Asset: Sci-Fi Soldier Character (Low Poly)

- Source: Sketchfab
<https://sketchfab.com/3d-models/scifi-soldier-character-low-poly-6c53691fa07e46aca890c771940b3651>
- Hermiston, P. (2025). *Sci-Fi Soldier Character (Low Poly)* [3D model]. Sketchfab.
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Asset: Sci-Fi Display Earth

- Source: Sketchfab
<https://sketchfab.com/3d-models/scifi-display-earth-821b2f02f584479bb8dfe5f61f51cccb>

- Sousinho. (2023). *Sci-Fi Display Earth* [3D model]. Sketchfab.
- License: Free Standard License (Sketchfab).
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Asset: Wires

- Source: Sketchfab
<https://sketchfab.com/3d-models/wires-fca671dcab294aff97fc422915e791ef>
- Bulat.Shakirov. (2022). *Wires* [3D model]. Sketchfab.
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Asset: Formula 1

- Source: Sketchfab
<https://sketchfab.com/3d-models/formula-1-lotus-49c-d08e707e8d4442469f3bfacd5d5d3441>
- dark_igorek. (2024). *Formula 1 Lotus 49C* [3D model]. Sketchfab.
- License: Licensed under Creative Commons Attribution (CC BY 4.0).
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Asset: Atomic Bomb Trinity Test

- Source: Sketchfab
<https://sketchfab.com/3d-models/atomic-bomb-trinity-nuclear-test-49205a3213964b83998b3263a180cf07>
- Sameer.Rayeen. (2024). *Atomic Bomb Trinity Nuclear Test* [3D model]. Sketchfab.
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Asset: NASA Super Heavy Rocket Project

- Source: Sketchfab
<https://sketchfab.com/3d-models/nexus-nasa-1960s-super-heavy-rocket-project-a5a60953a21b41f0943b5308d742752a>
- SebastianSosnowski. (2018). *NASA Super Heavy Rocket Project* [3D model]. Sketchfab.
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Asset: Rocket Engine

- Source: Sketchfab
<https://sketchfab.com/3d-models/rocket-engine-568bfa7ea4994541a65e63f8a6154838>
- Drift. (2024). *Rocket Engine* [3D model]. Sketchfab.

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Asset: Sci-Fi Terminal

- Source: Sketchfab
<https://sketchfab.com/3d-models/scifi-terminal-2-b03bde63cb4a42d29aff418888a38297>
- Sousinho. (2023). *Sci-Fi Terminal* [3D model]. Sketchfab.
- License: Free Standard License (Sketchfab).
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Asset: Mando Helmet

- Source: Sketchfab
<https://sketchfab.com/3d-models/mando-helmet-9da2e7fc67de443e95e190fc0c29f0c7>
- McCarthy3D. (2019). *Mando Helmet* [3D model]. Sketchfab.
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Asset: Tron Arcade Machine

- Source: Sketchfab
<https://sketchfab.com/3d-models/bally-midway-tron-arcade-machine-ada3f72316e4432aaadd043cdc3f257d>
- Big Jojo. (2025). *Bally Midway Tron Arcade Machine* [3D model]. Sketchfab.
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Asset: Police Drone

- Source: Sketchfab
<https://sketchfab.com/3d-models/police-drone-7f014be2705844c49152bfa1eb66fd86>
- Tom F. (2017). *Police Drone* [3D model]. Sketchfab.
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Asset: Drone

- Source: Sketchfab
<https://sketchfab.com/3d-models/drone-ad0e1d85295847409e961de33c2ee0bf>
- GreenG. (2021). *Drone* [3D model]. Sketchfab.
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Asset: Arc Reactor Concept Art

- Source: Sketchfab
<https://sketchfab.com/3d-models/arc-reactor-concept-art-f5a4da071b6d4c4aac529b8773ece3d2>
- Aryan Bajoria. (2025). Arc Reactor — Concept Art [3D model]. Sketchfab.
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